ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________Syzdykova Z.A.
«______» of ____________ 2024

Syllabus
Academic Year 2024 -2025

1. General Information	
Course title	Data visualization
Degree cycle (level)/ major	6B06101 "Computer Science"
Year, term	3, 1
Number of credits	5
Language of delivery:	English
Prerequisites	Basic understanding of Python programming language
Postrequisites	Machine learning
Lecturer(s)	Muratova Azhar, MSc in technical sciences, teacher azhar.muratova@astanait.edu.kz Astana IT University, Expo, C1 block, 2nd floor, office C1.1.329

2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	"Data visualization" is a 10-week course, which will equip students with the skills they need to advance in their computer science career. Data Visualization is the graphical representation of data. It involves producing efficient visual elements like charts, dashboards, graphs, mappings, etc. In this case python is a highly popular general-purpose programming language and it comes extremely useful for Data Scientists to create beautiful visualizations. Python provides the Data Scientists with various packages both for data processing and visualization.
2. Course Goal(s)	The purpose of this course is to introduce students to the broad set of techniques and job responsibilities associated with the role of a data scientist and to take a deep dive into the specific technique of data visualization by performing it in python. Not only for Data Visualization, but every process to be held in Python should also be started by importing the required packages. In this course we will use primary packages including Pandas for Data processing, Matplotlib for visuals, Seaborn for advanced visuals, and Numpy for scientific calculations.
3. Course Objectives:	• The learning objectives of a course are to help students understand the art of data representation using Python; • Students should be able to use various Python libraries like Matplotlib, Seaborn and Plotly for generating professional-grade visual aids of data insights • Gain the ability to analyze large data sets, perform data cleaning, wrangling, and manipulation using Python

4. Skills & Competencies	• presenting complex data in an easy-to-understand format • enhancing data analysis and decision-making • increases employability as proficiency in Python and data visualization are in high demand in many industries • equips students with knowledge to perform advanced statistical analyses
5. Course Learning Outcomes:	Students will have a good understanding of the main elements of data visualization such as: • handling, analyzing, and visualization data using Python libraries such as Matplotlib, Seaborn, and Pandas; • fostering strategic decision-making and data-driven solutions; • creating various types of plots, charts, and graphs to interpret complex data sets, and understand how to customize these visualizations; • learn the methodologies to present data that can facilitate informed decision making.
6. Methods of Assessment	• Weekly quizzes and assignments;
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: 1. Learning Python. Game programming, data visualization, web applications 2017- Eric Matiz. 2. Data Visualization in Python 2020 - Daniel Nelson 3. Lecture slides (available on moodle.astanait.edu.kz); Supplementary literature: 1. Python. Data Visualization 2020 - Abdrakhmanov M.I. 2. Python. Data Visualization . Matplotlib. Seaborn. Mayavi 2020 - Dev Practice Team
8. Resources	Online journals, articles, papers, books, internet resources, emulators, and simulation software.
9. Course policy	**Course and University policies include:** **Attendance:** Students are expected to attend all scheduled class sessions with all required reading and supplementary materials. Readings are to be completed before class. The student won't obtain additional points for course attendance, but attendance is important to pass the course, in particular, to pass Mid-term and End-Term control exams. **The total point of Attendance is**

	70% or more to pass every separate exam (Midterm, Endterm, and Final exams). In case the student is not able to attend the classes for some reasons, he/she must inform the teacher of the course or dean's office in advance and the student itself is responsible for self-learning all materials, which were given during unattended lessons. In case **the student did not attend more than 30% of the classes without any reasonable excuses** and did not inform in advance the dean's office or the teacher of the course then **the teacher has a right to mark him as "not graded", and a student wouldn't be admitted to the exam** (Midterm, Endterm, and Final exams). In other words, **students must participate in at least 70% of all class time, otherwise, he/she can't pass exams and can fail the course.** At the same time, in case **the student did not attend more than 30% of the classes without any reasonable excuses, or is not able to attend the classes for some reasons, the teacher can able to give themself student the opportunity to submit assignments, practical works and others tasks on the not studied topics during an additional time.** For example, teachers have the right to give an additional task, mini-projects or any other case studies in the framework of practical works or assignments on the studied course. **The evaluation and grading process should be done by the teacher and students before starting the Midterm Exam or the Endterm Exam.** This individual opportunity can be considered by the teacher in case the student can not present approved medical documents from medical polyclinics for any reason. In any other case, **students must attend at least 70% of all classes.** **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly graded, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Build on other comments. **Class work**: The duration of each lecture and the practical lesson is 50 minutes for offline classes and 40 minutes for online classes. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student misconduct, the student can be expelled from the classes.

	Being late to class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, being late to the class is not welcome and can have restriction activities by the teacher. **Attestation I and II:** Students with scores less than 25% for Attestation period I or Attestation period II (RK1/RK2) are automatically failed and should take the course again. **Homework / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In the case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will result in cheating from the students and may be the subject of additional disciplinary measures. **Late assignments:** Late assignments: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the teacher and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances of the teacher.* **Final exam* / Final project:** The final Examination will be on the LMS Moodle as a Final Quiz. **Laptops and mobile devices** can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the teacher. **Online lessons** can be used in case there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data;

	4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding the academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct codes will be posted to the students using the Learning Management System (moodle.astanait.edu.kz). **Contacting the Instructor (Teacher)**: The easiest and most reliable way to get in touch with the teacher is by email. Students must feel free to send an email if they have a question related to the course. The teachers will respond as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the teacher in their office during office hours to discuss the class using both offline and online ways.

3. Course Content

#	Abbreviation	Meaning
1	TSIS	Teacher-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction and general information about the course. Overview of tools: Google Sheets, MySQL, Power BI, Python (numpy, pandas, matplotlib libraries)	2	3	0	1	9
2	Google Sheets for data visualization. Using functions and formulas to prepare data for visualization. Creating interactive dashboards in Google Sheets.	2	3	0	1	9

3	Introduction to relational databases and SQL. SELECT queries and working with data. Aggregating data using GROUP BY and HAVING. Preparing data for visualization using SQL queries	2	3	0	1	9
4	Introduction to Power BI: interface and basic features. Import data from various sources (including MySQL). Building dashboards in Power BI.	2	3	0	1	9
5	Installing and configuring Python and the necessary libraries (numpy, pandas, matplotlib). Numpy Library. Numpy Library: Array Operations.	2	3	0	1	9
6	Pandas library: reading, writing and cleaning data. Working with DataFrame files. Example of using Pandas libraries.	2	3	0	1	9
7	Introduction to Matplotlib library for data visualization. Dataset analysis using Matplotlib in Python.	2	3	0	1	9
8	Mastering Data Visualization with Seaborn: In-Depth Techniques, Customization, and Real-World Applications	2	3	0	1	9
9	Introduction to data parsing. Working with API to receive data. Automate tasks using Apache Airflow. Version control and collaboration with GIT.	2	3	0	1	9
10	Integration of tools in one project. Applying all learned methods to solve a real problem. Presentation of projects	2	3	0	1	9
	Total hours: 150	**20**	**30**	**0**	**10**	**90**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	Work with lists	9	Books, internet resources	Labs
2	Branching operator if	9	Books, internet resources	Labs
3	Storing functions in modules	9	Books, internet resources	Labs
4	Introduction to classes	9	Books, internet resources	Labs
5	Exception handling	9	Books, internet resources	Labs
6	Scikit-learn library	9	Books, internet resources	Labs
7	Data science in banks	9	Books, internet resources	Labs
8	Tuples, dictionaries and sets	9	Books, internet resources	Labs
9	Python libraries	9	Books, internet resources	Labs
10	If, Else, While, for operators	9	Books, internet resources	Labs

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	Assignments**: Assignment 1 (week 1) Assignment 2 (week 2) Assignment 3 (week 3) **Midterm:**	**60** 20 20 20 **40**	100
2nd attestation	Assignments**: Assignment 6 (week 6) Assignment 7 (week 7) Assignment 8 (week 8) **End-term: Final project**	**60** 20 20 20 **40**	100
Final exam	**Final Exam***		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

** *The number of assignments can be different. It depends on the course program and is designed by the course syllabus. However, the total points of the assignment are 60 in each control period.*

* *Final exam can be given by the instructor in any other form such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, the following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught

25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must-pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), a total of 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **TSIS (Teacher Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by Instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and endterm exams is complex.
- **Final assessment** for the course "Model-Based Design" includes several questions and problems to be answered and solved. If the final examination will be online then at the completion of the exam, all works must be submitted as a single PDF file in the Learning Management System (moodle.astanait.edu.kz). No late submissions are allowed in the exam. If the final examination will be offline then after the exam, all works (hard copies of exam solution papers) must be given to the Instructor.

6. Lecturer (lecturers) approvals

Full name	**Job title**	**Date**	**Sign**
Azhar Muratova	Teacher	26/02/24	

Director **Sergaziyev M.Zh.**